STYLENOVA – DBMS WEEK 8

DATABASE RELATIONSHIP ANALYSIS USING JOINS

1. Customer and Orders – INNER JOIN

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         O.ORDER_ID,
  4         O.ORDER_DATE
  5  FROM CUSTOMERS C
  6  INNER JOIN ORDERS O
  7  ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
        201 Aarthi                                                    301
01-OCT-26

        202 Priya                                                     302
05-OCT-26

        203 Rahul                                                     303
10-OCT-26
```

2. Orders and Products – INNER JOIN

```
SQL> SELECT O.ORDER_ID,
  2         O.ORDER_DATE,
  3         P.PRODUCT_ID,
  4         P.PRODUCT_NAME,
  5         O.QUANTITY,
  6         P.PRICE
  7  FROM ORDERS O
  8  INNER JOIN PRODUCT P
  9  ON O.PRODUCT_ID = P.PRODUCT_ID;

  ORDER_ID ORDER_DAT PRODUCT_ID
---------- --------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
       302 05-OCT-26       102
Denim Jeans
         2       1299

       303 10-OCT-26       104
Kids Hoodie
         1        749

  ORDER_ID ORDER_DAT PRODUCT_ID
---------- --------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------

       301 01-OCT-26       101
Cotton T-Shirt
         1        599
```

3. **Product and Orders – INNER JOIN**

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         O.ORDER_ID,
  4         O.QUANTITY,
  5         P.PRICE,
  6         (O.QUANTITY * P.PRICE) AS SUBTOTAL
  7  FROM PRODUCT P
  8  INNER JOIN ORDERS O
  9  ON P.PRODUCT_ID = O.PRODUCT_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY      PRICE   SUBTOTAL
---------- ---------- ---------- ----------
       102
Denim Jeans
       302          2       1299       2598

       104
Kids Hoodie
       303          1        749        749

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY      PRICE   SUBTOTAL
---------- ---------- ---------- ----------

       101
Cotton T-Shirt
       301          1        599        599
```

4. **Display All Customers and Their Orders – LEFT JOIN**

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         O.ORDER_ID,
  4         O.ORDER_DATE,
  5         O.QUANTITY
  6  FROM CUSTOMERS C
  7  LEFT JOIN ORDERS O
  8  ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT   QUANTITY
--------- ----------
        201 Aarthi                                                    301
01-OCT-26          1

        202 Priya                                                     302
05-OCT-26          2

        203 Rahul                                                     303
10-OCT-26          1
```

5. **Display All Orders and Customer Details – RIGHT JOIN**

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         O.ORDER_ID,
  4         O.ORDER_DATE,
  5         O.QUANTITY
  6  FROM CUSTOMERS C
  7  RIGHT JOIN ORDERS O
  8  ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT   QUANTITY
--------- ----------
        201 Aarthi                                                    301
01-OCT-26          1

        202 Priya                                                     302
05-OCT-26          2

        203 Rahul                                                     303
10-OCT-26          1
```

6. **Complete Order Details**

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         O.ORDER_ID,
  4         O.ORDER_DATE,
  5         P.PRODUCT_ID,
  6         P.PRODUCT_NAME,
  7         O.QUANTITY,
  8         P.PRICE,
  9         (O.QUANTITY * P.PRICE) AS SUBTOTAL
 10  FROM CUSTOMERS C
 11  INNER JOIN ORDERS O
 12  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 13  INNER JOIN PRODUCT P
 14  ON O.PRODUCT_ID = P.PRODUCT_ID
 15  ORDER BY O.ORDER_DATE DESC;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_ID
--------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE   SUBTOTAL
---------- ---------- ----------
       203 Rahul                                                     303
10-OCT-26        104
Kids Hoodie
         1        749        749


CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_ID
--------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE   SUBTOTAL
---------- ---------- ----------
       202 Priya                                                     302
05-OCT-26        102
Denim Jeans
         2       1299       2598


CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_ID
--------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE   SUBTOTAL
---------- ---------- ----------
       201 Aarthi                                                    301
01-OCT-26        101
Cotton T-Shirt
         1        599        599
```

7. Customer Purchase History

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         O.ORDER_ID,
  4         O.ORDER_DATE,
  5         P.PRODUCT_NAME,
  6         O.QUANTITY,
  7         (O.QUANTITY * P.PRICE) AS SUBTOTAL
  8  FROM CUSTOMERS C
  9  INNER JOIN ORDERS O
 10  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 11  INNER JOIN PRODUCT P
 12  ON O.PRODUCT_ID = P.PRODUCT_ID
 13  ORDER BY C.CUSTOMER_ID, O.ORDER_DATE;

CUSTOMER_ID CUSTOMER_NAME                                    ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL
---------- ----------
         201 Aarthi                                                301
01-OCT-26
Cotton T-Shirt
         1        599


CUSTOMER_ID CUSTOMER_NAME                                    ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL
---------- ----------
         202 Priya                                                 302
05-OCT-26
Denim Jeans
         2       2598


CUSTOMER_ID CUSTOMER_NAME                                    ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL
---------- ----------
         203 Rahul                                                 303
10-OCT-26
Kids Hoodie
         1        749
```

8. Multi-Table Business Report

```
SQL> SELECT C.CUSTOMER_NAME,
  2         COUNT(DISTINCT O.ORDER_ID) AS TOTAL_ORDERS,
  3         SUM(O.QUANTITY * P.PRICE) AS TOTAL_PURCHASE,
  4         SUM(O.QUANTITY) AS TOTAL_PRODUCTS
  5  FROM CUSTOMERS C
  6  INNER JOIN ORDERS O
  7  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  INNER JOIN PRODUCT P
  9  ON O.PRODUCT_ID = P.PRODUCT_ID
 10  GROUP BY C.CUSTOMER_NAME
 11  ORDER BY TOTAL_PURCHASE DESC;

CUSTOMER_NAME                                    TOTAL_ORDERS TOTAL_PURCHASE
-------------------------------------------------- ------------ --------------
TOTAL_PRODUCTS
--------------
Priya                                                       1           2598
             2

Rahul                                                       1            749
             1

Aarthi                                                      1            599
             1
```